

we conclude that the columns of A^{-1} are linearly independent. $\square$

8) If C is 6×6 and the equation $Cx = v$ is consistent for every v in $\mathbb{R}^6$, is it possible that for some v , the equation $Cx = v$ has more than one solution? Why or why not?

Solution: By the defn. of consistency, for every v in $\mathbb{R}^6$, there ~~the~~ exists at least one solution for the equation $Cx = v$. By the Invertible Matrix Thm, and the equivalence of the statements (g) and (a), A is an invertible matrix. By Theorem 5, the equation $Cx = v$ has a unique solution $x = C^{-1}v$ for each v in $\mathbb{R}^6$. $\square$

19) If the columns of a 7×7 matrix D are linearly independent, what can you say about solutions of $Dx = b$? Why?

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (e) and (a), D is an invertible matrix. By Theorem 5, for each b in $\mathbb{R}^7$, the equation $Dx = b$ has a unique solution $x = D^{-1}b$. $\square$

20) If $n \times n$ matrices E and F have the property that $EF = I$, then E and F commute. Explain why.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (j) and (a), F is an invertible matrix. By defn, there exists a unique $n \times n$ matrix F^{-1} s.t. $FF^{-1} = F^{-1}F = I$. $\therefore F^{-1} = E$. So, $EF = FE = I$. $\therefore E$ and F commute. $\square$

21) If the equation $Gx = y$ has more than one solution for some y in $\mathbb{R}^n$, can the columns of G span $\mathbb{R}^n$?

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (a) and (h), we know that G is an invertible $n \times n$ matrix if and only if the columns of G span $\mathbb{R}^n$. By Theorem 5, for each $\hat{y}$ in $\mathbb{R}^n$, if G is invertible then the equation $Gx = y$ has the unique solution $x = G^{-1}y$. $\therefore$ For every y in $\mathbb{R}^n$, the equation $Gx = y$ has a unique solution if and only if the columns of G span $\mathbb{R}^n$.

$\therefore$ Since the equation $Gx = y$ has more than one solution for some y in $\mathbb{R}^n$, the columns of G do not span $\mathbb{R}^n$. $\square$